Chapter 4 • Factorization and Algebraic Manipulation

Solved Exercise 4.3

1. Find HCF by factorization.

(i) $21x^2y=7xy\cdot 3x$ and $35xy^2=7xy\cdot 5y \Rightarrow$ HCF$=7xy$.
(ii) $4x^2-9y^2=(2x-3y)(2x+3y)$, $2x^2-3xy=x(2x-3y)$
$\Rightarrow$ HCF$=2x-3y$.
(iii) $x^3-1=(x-1)(x^2+x+1)$
$\Rightarrow$ HCF$=x^2+x+1$.
(iv) $a^3+2a^2-3a=a(a+3)(a-1)$
$2a^3+5a^2-3a=a(a+3)(2a-1)$
$\Rightarrow$ HCF$=a(a+3)$.
(v) factors contain respectively $(t-4)(t+1)$, $(t+4)(t+1)$, $(t-1)(t+1) \Rightarrow$ HCF$=t+1$.
(vi) factors contain $(x+7)(x+8)$, $(x+8)(x-3)$, $x(x+8) \Rightarrow$ HCF$=x+8$.

2. Find HCF by the division method.

(i) $27x^3+9x^2-3x-10=(3x-2)(9x^2+9x+5)$, so division ends with zero remainder: HCF$=3x-2$.
(ii) $x^3-9x^2+23x-15 = (x^2-4x+3)(x-5)$: HCF
$= x^2-4x+3$.
(iii) $2x^3+2x^2+2x+2=2(x+1)(x^2+1)$; $6x^3+12x^2+6x+12=6(x+2)(x^2+1)$. Their shared factors give HCF$=2(x^2+1)$.
(iv) Direct verification of the terminal common divisor: $2x^3-4x^2-16x=2x(x-4)(x+2)$, $x^3-4x=x(x-2)(x+2)$, $3x^2-6x=3x(x-2)$.
x , not $x(x-2)$.

3. Find LCM by prime factorization.

(i) LCM$(2a^2b,4ab^2,6ab)=12a^2b^2$.
(ii) $x^2+x=x(x+1)$, $x^3+x^2=x^2(x+1)$
$\Rightarrow$ LCM$=x^2(x+1)$.
$x^2(x+1)$
(iii) $a^2-4a+4=(a-2)^2$, $a^2-2a=a(a-2)$
$\Rightarrow$ LCM$=a(a-2)^2$.
(iv) x^4-16 and $x^3-4x=x(x-2)(x+2) \Rightarrow$ LCM$=x(x^4-16)$.
(v) $16-4x^2=4(4-x^2)$, $x^2+x-6=(x+3)(x-2)$, and $4-x^2=-(x-2)(x+2) \Rightarrow$ LCM$=4(4-x^2)(x+3)$, up to sign.
$4(4-x^2)(x+3)$.

4. HCF$=y-7$, LCM$=y^3-10y^2+11y+70$, one polynomial$=y^2-5y-14$. Find the other.

LCM$=(y-7)(y-5)(y+2)$, and the given polynomial$=(y-7)(y+2)$.
Other = HCF$\times$LCM/given $= (y-7)(y-5)$.
$y^2-12y+35$.

5. LCM$=36x^3(x+a)(x^3-a^3)$, HCF$=x^2(x-a)$, $p(x)=4x^2(x^2-a^2)$. Find $q(x)$.

$q=\text{LCM}\times\frac{\text{HCF}}{p}$
Cancel x^2, $(x-a)$, $(x+a)$, and the factor 4.
$q(x)=9x^3(x^3-a^3)$.

6. HCF$=x+a$ and LCM$=12x^2(x+a)(x^2-a^2)$. Find the product of the polynomials.

Product = HCF$\times$LCM
$= 12x^2(x+a)^2(x^2-a^2)$.
$12x^2(x-a)(x+a)^3$.

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